I- Introduction

Definition

If $f(x)$ is as close as we like to some real number A for all x sufficiently close to (but not equal to) a, then we say that $f(x)$ has a limit of A as x approaches a, and we write $\lim_{x\to a} f(x) = A$.

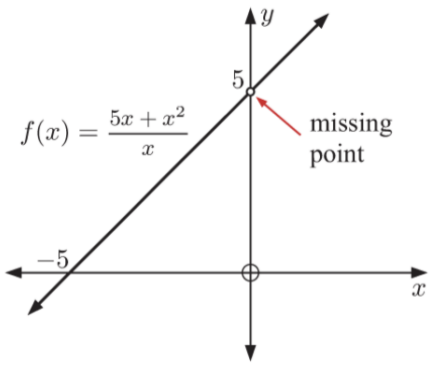
For example, suppose we wish to find the limit of $f(x) = \dfrac{5x + x^2}{x}$ as $x \to 0$.

It is tempting for us to simply substitute $x = 0$ into $f(x)$. However, in doing this, not only do we get the meaningless value of $\frac{0}{0}$, but also we ignore the basic limit definition.

Instead, observe that if $f(x) = \dfrac{5x + x^2}{x} = \dfrac{x(5 + x)}{x}$

$$\text{then} \quad f(x) = \begin{cases} 5 + x & \text{if } x \neq 0 \\ \text{is undefined if } x = 0. \end{cases}$$

The graph of $y = f(x)$ is therefore the straight line $y = x+5$ with a "hole" at $(0, 5)$. The "hole" is a **point of discontinuity** of the function.



However, even though this point is missing, the *limit* of $f(x)$ as x approaches 0 does exist. In particular, as $x \to 0$ from either direction, $f(x) \to 5$.

We write $\lim\limits_{x\to 0} \dfrac{5x + x^2}{x} = 5$ which reads:

"the limit as x approaches 0, of $\dfrac{5x + x^2}{x}$, is 5".

Property

Consider functions $f(x)$ and $g(x)$ for which $\lim_{x\to a} f(x) = l$ and $\lim_{x\to a} g(x) = m$, where a, l, $m \in \mathbb{R}$.

- $\lim_{x\to a}(f(x) \pm g(x)) = l \pm m$
- $\lim_{x\to a} f(x)\, g(x) = lm$
- $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{l}{m}$ provided $m \neq 0$

Example

Evaluate :

1. $\lim_{x\to 2} x^2$
2. $\lim_{x\to 0} \dfrac{x^2 + 3x}{x}$
3. $\lim_{x\to 3} \dfrac{x^2 - 9}{x - 3}$

EXERCISE 1

Evaluate:

a $\lim\limits_{x\to 1} \dfrac{x^2-3x}{x}$ b $\lim\limits_{h\to 2} \dfrac{h^2+5h}{h}$ c $\lim\limits_{x\to 0} \dfrac{x-1}{x+1}$

EXERCISE 2

Evaluate:

a $\lim\limits_{x\to 3} (x+4)$ b $\lim\limits_{x\to -1} (5-2x)$ c $\lim\limits_{x\to 4} (3x-1)$

d $\lim\limits_{x\to 2} (5x^2-3x+2)$ e $\lim\limits_{h\to 0} h^2(1-h)$ f $\lim\limits_{x\to 0} (x^2+5)$

EXERCISE 3

Explain why $\lim\limits_{x\to 0} \dfrac{x}{x} = 1$, even though $\dfrac{0}{0}$ is meaningless.

EXERCISE 4

Evaluate:

a $\lim\limits_{x\to 0} \dfrac{x^2-3x}{x}$ b $\lim\limits_{x\to 0} \dfrac{x^2+5x}{x}$ c $\lim\limits_{x\to 0} \dfrac{2x^2-x}{x}$

d $\lim\limits_{h\to 0} \dfrac{2h^2+6h}{h}$ e $\lim\limits_{h\to 0} \dfrac{3h^2-4h}{h}$ f $\lim\limits_{h\to 0} \dfrac{h^3-8h}{h}$

g $\lim\limits_{x\to 1} \dfrac{x^2-x}{x-1}$ h $\lim\limits_{x\to 2} \dfrac{x^2-2x}{x-2}$ i $\lim\limits_{x\to 3} \dfrac{x^2-x-6}{x-3}$

So far we have considered cases where the function $f(x)$ approaches some finite value A as x approaches a from either side. More formally, we can write this using the one-sided limits $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$.

Definition

The limit $\lim_{x\to a} f(x)$ exists and is equal to the finite value A if $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = A$. We say that $f(x)$ converges to A as x approaches a.

Remark. *We cannot write* $\lim_{x\to 0} f(x) = \infty$ *because* ∞ *is not a number. Rather,* $\lim_{x\to 0} f(x)$ *does not exist.*

Example

Find, if possible :

1. $\lim_{x\to 1} \dfrac{1}{x}$
2. $\lim_{x\to 0} \dfrac{1}{x}$

EXERCISE 5

a Sketch the graph of $y = -\dfrac{1}{(x-3)^2}$.

b Find, if possible:

i $\lim\limits_{x\to 0}\left(-\dfrac{1}{(x-3)^2}\right)$ ii $\lim\limits_{x\to 3}\left(-\dfrac{1}{(x-3)^2}\right)$ iii $\lim\limits_{x\to 5}\left(-\dfrac{1}{(x-3)^2}\right)$

EXERCISE 6

Find, if possible: a $\lim\limits_{x\to -1}(3x+2)$ b $\lim\limits_{x\to -2}\dfrac{x^2+x-2}{x-2}$

EXERCISE 7

Consider the function $f(x) = \dfrac{x}{x^2-2x}$.

a For what values of x is $f(x)$ undefined?

b Sketch the graph of $y = f(x)$.

c Find, if possible: i $\lim\limits_{x\to 2} f(x)$ ii $\lim\limits_{x\to 1} f(x)$ iii $\lim\limits_{x\to 0} f(x)$

EXERCISE 8

Let $f(x) = \begin{cases} x^2, & x > 3 \\ 5, & x = 3 \\ 3x, & x < 3. \end{cases}$

a Sketch the graph of $y = f(x)$.

b Evaluate $\lim\limits_{x\to 3^-} f(x)$ and $\lim\limits_{x\to 3^+} f(x)$.

c Does $\lim\limits_{x\to 3} f(x)$ exist? Explain your answer.

EXERCISE 9

Given $f(x) = \begin{cases} \sin x, & x < 0 \\ \dfrac{x^2-3x+5}{5}, & x > 0 \end{cases}$, discuss:

a $\lim\limits_{x\to 0^-} f(x)$ b $\lim\limits_{x\to 0^+} f(x)$ c $\lim\limits_{x\to 0} f(x)$

EXERCISE 10

Find, if possible:

a $\lim\limits_{x\to 0}\dfrac{\sin x}{e^x}$ b $\lim\limits_{x\to \pi}\dfrac{\sin x}{1-\cos x}$ c $\lim\limits_{\theta\to 0}\dfrac{\cos\theta}{\theta}$ d $\lim\limits_{x\to 2}\dfrac{\ln x}{\sqrt{2-x}}$

EXERCISE 11

Let $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{otherwise.} \end{cases}$

a Find: i $f(\frac{1}{2})$ ii $f(\sqrt{2})$ iii $f(0)$

b Find, if possible: i $\lim\limits_{x\to \frac{1}{2}} f(x)$ ii $\lim\limits_{x\to \sqrt{2}} f(x)$ iii $\lim\limits_{x\to 0} f(x)$

Explain your answers.

II- Specific limits

a) Infinity

Property

$$\lim_{x\to\infty} \frac{1}{x^n} = 0 \quad \text{for all } n > 0.$$

$$\lim_{x\to-\infty} \frac{1}{x^n} = 0 \quad \text{for all } n \in \mathbb{Z}^+.$$

Proof

Notice that as $x \to \infty$, $1 < x < x^2 < x^3 < \ldots$ and as x gets very large, the value of $\frac{1}{x}$ gets very small. In fact, we can make $\frac{1}{x}$ as close to 0 as we like by making x large enough. This means that $\lim_{x\to\infty} \frac{1}{x} = 0$, even though $\frac{1}{x}$ never actually reaches 0.

Example

Evaluate :

1. $\lim_{x\to\infty} \frac{2x+3}{x-4}$
2. $\lim_{x\to\infty} \frac{x^2-3x+2}{1-x^2}$
3. $\lim_{x\to\infty} \frac{x^2+x+1}{x-2}$

EXERCISE 12

Evaluate:

a $\lim_{x\to\infty} \frac{3x-2}{x+1}$ b $\lim_{x\to\infty} \frac{1-2x}{3x+2}$ c $\lim_{x\to\infty} \frac{x}{1-x}$

d $\lim_{x\to\infty} \frac{x^2+3}{x^2-1}$ e $\lim_{x\to\infty} \frac{x^2-2x+4}{4x^2-x-1}$ f $\lim_{x\to\infty} \frac{x+4}{2x^2+x-1}$

g $\lim_{x\to\infty} \frac{x+5}{x^2-x+4}$ h $\lim_{x\to\infty} \frac{x^2+3x-4}{x-1}$ i $\lim_{x\to\infty} \frac{2x^2-x+1}{5x+3}$

Example

1. Discuss the behaviour of $f(x) = \frac{2-x}{1+x}$ near its asymptotes, and hence deduce their equations.
2. If they exist, state the values of $\lim_{x\to-\infty} f(x)$ and $\lim_{x\to\infty} f(x)$.

b) Trigonometry